

DILEEP KUMAR THIRUVENKADAM

M.Sc. Software Engineering Student | Associate Software Developer



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DOB: 06.11.2001 | **Work Permit:** Student Visa (140 full / 280 half days)

PROFESSIONAL SUMMARY

Software Engineering Master's student at Hochschule Heilbronn with a focus on DevSecOps, Cloud Infrastructure, and Automation. With professional experience in backend development and test orchestration, I bridge the gap between development and operations. My academic performance includes a 1.0 grade in Cloud Computing and DevOps & SecOps. I am proficient in Python, Java, and JavaScript frameworks.

CORE TECHNICAL SKILLS

Cloud & Infrastructure:	AWS (S3, CloudWatch, SNS, IAM), Kubernetes (EKS), Docker, Terraform (Infrastructure as Code), GitHub Actions.
DevOps & Security:	CI/CD Pipeline Design, SonarQube (Static Analysis), Vulnerability Scanning, Shift-Left Security.
Software Development:	Python (Advanced), Java (Spring Boot), JavaScript (Node.js, Express, ReactJS), SQL (PostgreSQL, MySQL), Bash Scripting.
QA & Testing:	Selenium, Cypress, PyTest, RestAssured (API Testing), JUnit, Regression Testing, Manual UAT, Jira.
Data & Analytics:	Snowflake, Databricks, Pandas, NumPy, Scikit-learn, Power BI.

PROFESSIONAL EXPERIENCE

Associate Software Developer

2022 – 2024

Quaking Aspen Private Limited, India

- Backend Engineering:** Developed and optimized core backend modules on AWS using Java and Python, resulting in a 15% reduction in system latency through improved query logic.
- Automation Leadership:** Architected comprehensive automated regression suites using Selenium and Cypress, decreasing manual testing efforts by 40% and increasing deployment frequency.
- API Development:** Engineered and documented RESTful APIs to facilitate seamless integration between front-end services and PostgreSQL databases.
- Agile Collaboration:** Actively participated in all Scrum ceremonies; managed lifecycle tasks using Jira and maintained version control integrity via Git.

Web Development Intern

2021

Zoho Corporation, India

- Frontend Optimization:** Designed and implemented responsive UI components using ReactJS, focusing on efficient state management and API data rendering.
- Testing:** Performed unit and integration testing to ensure cross-browser compatibility and UI consistency.

ACADEMIC PROJECTS

DevSecOps & Security Orchestration Pipeline (HHN Graduate Project)

- Designed a high-security CI/CD pipeline using GitHub Actions and Docker.
- Integrated SonarQube for automated code quality gates and security scanning, ensuring zero critical vulnerabilities in the production build.

Kubernetes Orchestration & Cluster Management

- Provisioned and managed microservices on Kubernetes clusters, implementing auto-scaling and load balancing to ensure 99.9% availability.
- Utilized Terraform for automated resource provisioning (IaC) to reduce environment setup time by 70%.

Intelligent Traffic Simulation System

- Developed a traffic flow modeling tool in Python to simulate real-world congestion scenarios and optimize urban routing algorithms.

EDUCATION

M.Sc. Software Engineering and Management

Sept 2024 – Present

Hochschule Heilbronn, Germany

- Current Average Grade: 2.2
- Notable Achievements: Grade 1.0 in Cloud Computing and DevOps & SecOps; Grade 1.3 in Advanced Software Architecture and Data Science Lab.

B.Tech. Computer Science and Engineering

2018 – 2022

SRM Institute of Science and Technology, India | Final Grade: 1.9 (German Scale)

CERTIFICATIONS

AWS Academy Cloud Developing

Google Data Analytics Professional Certificate

LANGUAGES

English: Fluent (C1 level) | **German:** A2 Level (Currently enrolled in intensive courses) | **Tamil:** Native